# 1 Forward kinematics models

## 1.1 DH description

The six revolute joints are described using the *standard* DH convention. Frame $\{0\}$ coincides with `base_link`, and $q_i = 0$ denotes the URDF home configuration. The endpoint $E$ is `link_6`. For each step $i$, the transform from frame $\{i-1\}$ to frame $\{i\}$ is

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i), \qquad \theta_i = q_i + \theta_i^0. \tag{1}$$

Joint $i$ rotates about $z_{i-1}$. For $i < 6$, the new axis $z_i$ lies on joint $i+1$; the terminal axis $z_6$ is chosen parallel to $z_5$. DH frame numbers therefore differ from physical joint numbers.

**How to read one step.** Choose $x_i$ along a common normal to $z_{i-1}$ and $z_i$. Its intersection with $z_i$ defines $O_i$. If two distinct axes intersect, their common-normal length is zero and $x_i$ is perpendicular to both axes through their intersection. For coincident axes, the origin and transverse $x$ axis are free; here the last origin is the endpoint. Set $y_i = z_i \times x_i$ to complete a right-handed frame. The four parameters then have the following geometric meaning:

| Parameter | Measured about / along | Construction from $\{i-1\}$ to $\{i\}$ |
|---|---|---|
| $\theta_i$ | about $z_{i-1}$ | Turn $x_{i-1}$ towards $x_i$. |
| $d_i$ | along $z_{i-1}$ | Move from $O_{i-1}$ to the foot of the common normal. |
| $a_i$ | along $x_i$ | Follow the common normal to $O_i$. |
| $\alpha_i$ | about $x_i$ | Turn $z_{i-1}$ towards $z_i$. |

Translations are signed along the indicated positive axes; rotations follow the right-hand rule. Figure 2 shows the construction at $q = 0$.

**Table 1:** Six DH steps. Lengths are in metres; angular constants are in radians.

| $i$ | $d_i$ | $\theta_i$ | $\alpha_i$ | $a_i$ | Geometric construction at home |
|---|---|---|---|---|---|
| 1 | 1 | $q_1$ | $-\pi/3$ | 0.8 | Rise $0.5+0.5$ from the base, then move 0.8 along $x_1$ to joint 2. Its axis gives a $-60°$ twist. |
| 2 | 1.25 | $q_2$ | $+\pi/2$ | 2 | Move 1.25 along joint 2's axis and 2 along $x_2$ to joint 3; twist by $+90°$. |
| 3 | 0.75 | $q_3 + \pi/2$ | $+\pi/2$ | 0 | Extend joints 3 and 4's axes to their intersection $O_3$. Turn $x_2$ by $+90°$ to select $x_3$; no common-normal translation is needed. |
| 4 | 2.5 | $q_4$ | $-\pi/2$ | 0 | From $O_3$, follow joint 4's axis by $1.5+1$ to the wrist centre $O_4$; twist by $-90°$. |
| 5 | 0 | $q_5$ | $+\pi/2$ | 0 | Joints 5 and 6's axes meet at the wrist centre. Keep $O_5 = O_4$ and twist by $+90°$. |
| 6 | 1.5 | $q_6$ | 0 | 0 | Follow joint 6's axis from the wrist centre to the endpoint. Choose $z_6 = z_5$ and no fixed twist. |

**Frame origins versus joint centres.** A DH origin lies on the selected joint axis but need not coincide with a URDF joint origin. In particular, $O_3$ lies 1.5 m before the

physical centre of joint 4. This distance enters $d_4$, giving $d_4 = 1.5 + 1 = 2.5$ m, while $a_3 = 0$. Similarly, $d_5 = 0$ even though the centres of joints 5 and 6 are 1.5 m apart: the two axes already intersect at $O_4 = O_5$.

**Endpoint alignment.** Frame $\{6\}$ and the URDF endpoint have the same origin but different axes: $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. Hence the forward kinematics uses the fixed alignment on the right:

$$ {}^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2} $$

## 1.2 Product of exponentials model

The product of exponentials (PoE) describes the robot through its joint axes at the home configuration. All vectors below are expressed in `base_link`, with $q = 0$ and endpoint `link_6` as in the DH model.

**1. Describe each revolute joint by a screw axis.** Let $\omega_i$ be the unit direction of joint $i$ and let $r_i$ be a point on its axis at home. With unit joint speed, a point $p$ has velocity $\omega_i \times (p - r_i) = \omega_i \times p + v_i$. This gives the six-vector $\xi_i$ and its matrix form:

$$ v_i = -\omega_i \times r_i, \qquad \xi_i = \begin{pmatrix} \omega_i \\ v_i \end{pmatrix}, \qquad \widehat{\xi}_i = \begin{pmatrix} [\omega_i]_\times & v_i \\ 0^T & 0 \end{pmatrix}. \tag{3} $$

Here $[\omega]_\times$ represents the cross product:

$$ [\omega]_\times = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}, \qquad [\omega]_\times p = \omega \times p. \tag{4} $$

Any point on the same axis gives the same $v_i$: adding a multiple of $\omega_i$ to $r_i$ does not change the cross product.

**2. Find the six axes at home.** The fixed tilt $\gamma = \pi/6$ maps the local $y$ and $z$ axes to $(0, \sqrt{3}/2, 1/2)^T$ and $(0, -1/2, \sqrt{3}/2)^T$. The points $r_i$ follow by adding the URDF joint offsets in the base frame. For example, $r_3 = r_2 + (2, 0, 0)^T + (5/4)\omega_2$. Applying (3) then gives Table 2.

At home, axes 4 and 6 lie on the same line and point in the same direction. Their chosen origins differ only along that line, so $\xi_4 = \xi_6$. This equality describes the home configuration; the axes need not remain aligned when joint 5 moves.

**3. Obtain the motion of one joint.** For a unit axis $\omega_i$, Rodrigues' formula gives the rotation $R_i(q_i)$. The corresponding rigid motion is the matrix exponential

$$ \begin{aligned} R_i(q_i) &= I_3 + \sin q_i [\omega_i]_\times + (1 - \cos q_i)[\omega_i]_\times^2, \\ e^{\widehat{\xi}_i q_i} &= \begin{pmatrix} R_i(q_i) & (I_3 - R_i(q_i))r_i \\ 0^T & 1 \end{pmatrix}. \end{aligned} \tag{5} $$

**Table 2:** Home screw data. Each vector is written as a row; $r_i$ uses the physical joint origin. Lengths are in metres.

| $i$ | $\omega_i^T$ | $r_i^T$ | $v_i^T$ |
|---|---|---|---|
| 1 | $(0, 0, 1)$ | $(0, 0, \frac{1}{2})$ | $(0, 0, 0)$ |
| 2 | $(0, \frac{\sqrt{3}}{2}, \frac{1}{2})$ | $(\frac{4}{5}, 0, 1)$ | $(-\frac{\sqrt{3}}{2}, -\frac{2}{5}, \frac{2\sqrt{3}}{5})$ |
| 3 | $(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$ | $(\frac{14}{5}, \frac{5\sqrt{3}}{8}, \frac{13}{8})$ | $(\frac{7}{4}, -\frac{7\sqrt{3}}{5}, -\frac{7}{5})$ |
| 4 | $(1, 0, 0)$ | $(\frac{43}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$ | $(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$ |
| 5 | $(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$ | $(\frac{53}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$ | $(\frac{7}{4}, -\frac{53\sqrt{3}}{20}, -\frac{53}{20})$ |
| 6 | $(1, 0, 0)$ | $(\frac{34}{5}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$ | $(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$ |

The translation term follows by writing the rotated point as $r_i + R_i(q_i)(p - r_i)$. It keeps every point on the joint axis fixed.

**4. Set the home pose and combine the joint motions.** At $q = 0$, the endpoint has orientation $R_x(\gamma)$ and position $r_6$. Its home pose is therefore

$$M = {}^0T_E(0) = \begin{pmatrix} 1 & 0 & 0 & \frac{34}{5} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} & \frac{5\sqrt{3}-3}{8} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{13+3\sqrt{3}}{8} \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{6}$$

The complete forward kinematics is

$$\boxed{{}^0T_E(q) = e^{\widehat{\xi}_1 q_1} e^{\widehat{\xi}_2 q_2} \cdots e^{\widehat{\xi}_6 q_6} M.} \tag{7}$$

The factors are written in joint order, from 1 to 6. The screw axes are fixed at their home values; the product accounts for how earlier joints move the later axes. At $q = 0$, all exponentials are identity matrices and the expression returns $M$.

### 1.3 Verification

Using the data above, the PoE, DH and direct URDF models were compared at 514 configurations: home, each joint alone at $\pm\pi/2$, one mixed pose, and 500 random configurations in $[-\pi, \pi]^6$. Across all model pairs, the largest position residual was $5.11 \times 10^{-11}$ m and the largest rotation-matrix Frobenius residual was $1.18 \times 10^{-11}$. The DH–PoE residuals were below $10^{-14}$; the URDF stores the fixed tilt as a decimal approximation of $\pi/6$.

## 2 Jacobian and singularities

**Construction from the PoE model.** The MATLAB PoE routine keeps the intermediate transforms. These give each joint's current unit direction $w_i$ and a point $p_i$ on its

axis, both in `base_link`. For a revolute joint, the Jacobian column is

$$ {}^{0}J_k(:,i) = \begin{pmatrix} w_i \times (p_k - p_i) \\ w_i \end{pmatrix}, \qquad i = 1, \ldots, 6. \tag{8} $$

The upper rows give linear velocity and the lower rows give angular velocity. We use $k = 6$ for the endpoint. Choosing $k = 5$ instead refers the same tool motion to the wrist centre.

**Wrist Jacobian.** Here ${}^{3}J_5$ denotes this full Jacobian at the wrist centre, expressed along the DH frame-3 axes. Both velocity blocks are rotated by $({}^{0}R_3)^T$; the velocities remain relative to the base. All three wrist axes pass through $p_5$, so their linear columns vanish. With $c_i = \cos q_i$ and $s_i = \sin q_i$, the result has the form

$$ {}^{3}J_5 = \begin{pmatrix} A & 0 \\ C & B \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -s_4 & c_4 s_5 \\ 0 & c_4 & s_4 s_5 \\ 1 & 0 & c_5 \end{pmatrix}. \tag{9} $$

The last three columns describe the wrist subchain. The blocks $A$ and $C$ give the linear and angular contributions of joints 1–3; $B$ gives the wrist's angular contribution.

**Symbolic and numerical routes.** The same MATLAB function accepts numerical or symbolic inputs. Numerical inputs evaluate the Jacobian at a chosen configuration. Symbolic inputs give an exact expression, from which the determinant is obtained. We use $J = {}^{3}J_5$, whose zero block makes the determinant easy to evaluate:

$$ \det J(q) = \sin q_5 \, D(q_2, q_3), \qquad D(q_2, q_3) = -\det A. \tag{10} $$

Changing the expression frame or velocity reference point preserves this determinant, provided all six columns describe the same full tool motion. The wrist factor vanishes at $q_5 = 0$ or $\pi$ modulo $2\pi$. Axes 4 and 6 then lie on the same line, so one independent rotation is lost. The arm factor depends jointly on $q_2$ and $q_3$. Thus the wrist and arm conditions separate, while the arm condition must be studied in the $(q_2, q_3)$ plane, as shown in Figure 1.

A full configuration is regular exactly when both factors are nonzero. For numerical checks, the smallest singular value of the full $J$ gives a measure of proximity to a singularity.

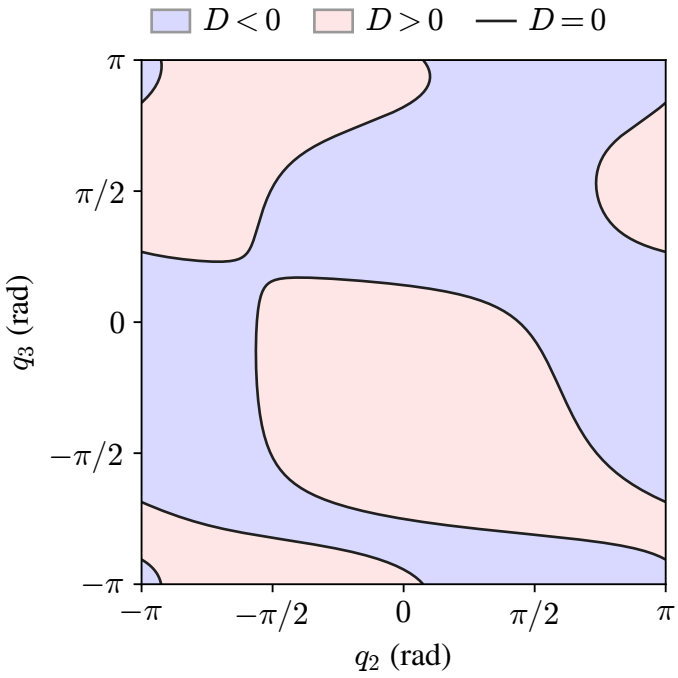



**Figure 1:** Arm singularity map over $(q_2, q_3) \in [-\pi, \pi]^2$. Black curves mark $D = 0$; the colours show its sign. Setting $q_5 = \pi/2$ gives $\det J = D$. Wrist singularities $\sin q_5 = 0$ are a separate condition. Opposite edges match modulo $2\pi$.

**How the intermediate PoE transforms give the axes.** Write the product before joint $i$ as $F_{i-1} = e^{\widehat{\xi}_1 q_1} \cdots e^{\widehat{\xi}_{i-1} q_{i-1}} = (R_{i-1}, t_{i-1})$. Then $w_i = R_{i-1}\omega_i$ and $p_i = R_{i-1} r_i + t_{i-1}$. The current screw's linear component is $-w_i \times p_i$. Evaluating its velocity field at $p_k$ adds $w_i \times p_k$, giving the upper block of (8).

**Why the determinant separates.** The zero block in (9) gives $\det({}^3J_5) = \det A \det B$, with $\det B = -\sin q_5$. Expressing both velocity blocks in another rotated frame multiplies the Jacobian by $\mathrm{diag}(R^T, R^T)$, whose determinant is one. If $d = p_\ell - p_j$, changing the velocity reference gives

$$ {}^0J_\ell = \begin{pmatrix} I_3 & -[d]_\times \\ 0 & I_3 \end{pmatrix} {}^0J_j. \tag{11} $$

This matrix also has determinant one. Thus $\det({}^iJ_j) = \det({}^3J_5)$ for every expression frame $i$ and velocity reference point $p_j$, when all six columns describe the same full tool motion. This is why ${}^3J_5$ suffices to study the full robot's singularities. Changing the reference point preserves rank, but can change the singular values. Numerical singular-value margins should therefore use a fixed reference point.

**Explicit arm factor.** Using the DH lengths and $\gamma = \pi/6$, define

$$ U = a_2 + d_4 c_3, \qquad X = U c_2 + d_3 s_2, \qquad Z = -U s_2 + d_3 c_2. \tag{12} $$

The symbolic result can then be written as

$$ D(q_2, q_3) = \frac{d_4}{2} \left[ a_1 Z c_3 - \sqrt{3}\, a_1 U s_3 - \sqrt{3}\, X (a_2 s_3 - d_2 c_3) \right]. \tag{13} $$

This expression contains neither $q_1$, $q_4$ nor $q_6$. Its zeros generally depend on the pair $(q_2, q_3)$, so the arm singularities are studied as curves in that plane.

# 3 Analytical inverse kinematics

The axes of joints 4–6 intersect at the wrist centre, allowing position and orientation to be solved separately. We first determine the arm angles $(q_1, q_2, q_3)$ from the desired wrist-centre position, then recover the wrist angles $(q_4, q_5, q_6)$ from the desired orientation.

## 3.1 Wrist-centre positioning

Let $R_d$ and $p_d$ denote the desired orientation and position of `link_6` relative to `base_link`. Since the endpoint offset lies along the tool's $x$ axis, the target wrist centre is

$$p_w = p_d - d_6 R_d e_x = (x_w, y_w, z_w)^T, \qquad e_x = (1, 0, 0)^T. \tag{14}$$

Throughout, $c_i = \cos q_i$ and $s_i = \sin q_i$.

### Derivation

**1. Relate the target wrist position to the three arm angles.** For a candidate $q_1$, undo the base rotation, translate to the shoulder at $s = (a_1, 0, d_1)^T$, and undo the fixed tilt $\gamma = \pi/6$. The resulting auxiliary frame follows the base rotation and fixed tilt, but not the joint-2 rotation. With $h = z_w - d_1$ and $(c_\gamma, s_\gamma) = (\sqrt{3}/2, 1/2)$,

$$\begin{aligned} r(q_1) &:= R_x(-\gamma)\Big(R_z(-q_1)p_w - s\Big) = (X, Y, Z)^T, \\ X &= c_1 x_w + s_1 y_w - a_1, \\ Y &= c_\gamma(-s_1 x_w + c_1 y_w) + s_\gamma h, \\ Z &= -s_\gamma(-s_1 x_w + c_1 y_w) + c_\gamma h. \end{aligned} \tag{15}$$

These coordinates still depend on the unknown $q_1$.

In the moving URDF frame of `link_2`, the shoulder-to-elbow vector is $a = (a_2, d_2, 0)^T$. The elbow-to-wrist vector is $R_z(q_3)b_0$, where $b_0 = (d_4, 0, d_3)^T$ is expressed in the URDF frame of `link_3_d3`. Applying the shoulder rotation maps their sum into the auxiliary frame:

$$r(q_1) = R_y(q_2)\Big(a + R_z(q_3)b_0\Big) = R_y(q_2)\begin{pmatrix} a_2 + d_4 c_3 \\ d_2 + d_4 s_3 \\ d_3 \end{pmatrix}. \tag{16}$$

The rotation $R_z(q_3)$ above is expressed using the URDF link frames, with $q_3 = 0$ at the home configuration. The DH frames use a different reference orientation: at home, their angular parameter is already $\theta_3 = \pi/2$. Consequently, $\theta_3 = q_3 + \pi/2$, whereas the rotation in this geometric derivation uses $q_3$ directly.

**2. Express $\sin q_3$ and $\cos q_3$ as functions of $q_1$.** Rotation about $y$ preserves both the $y$ component and the norm. Equation (16) therefore gives two relations independent of $q_2$:

$$\begin{aligned} Y &= d_2 + d_4 s_3, \\ N := X^2 + Y^2 + Z^2 &= a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 d_4 s_3. \end{aligned} \tag{17}$$

Setting $K = d_2^2 - a_2^2 - d_3^2 - d_4^2$ yields

$$s_3 = \frac{Y - d_2}{d_4}, \qquad c_3 = \frac{N + K - 2d_2 Y}{2a_2 d_4}. \tag{18}$$

Here $a_2 = 2$ and $d_4 = 2.5$ are nonzero. For a trial value of $q_1$, these expressions give the required values of $\sin q_3$ and $\cos q_3$. They correspond to a real elbow angle only if $s_3^2 + c_3^2 = 1$. This consistency condition provides the equation used to determine $q_1$.

**3. Determine the candidate base angles $q_1$.** Since $R_x(-\gamma)$ preserves the norm, $N = \|r(q_1)\|^2 = \|R_z(-q_1)p_w - s\|^2$. Defining $N_0 = x_w^2 + y_w^2 + h^2 + a_1^2$ gives the compact expression

$$N = N_0 - 2a_1(x_w c_1 + y_w s_1). \tag{19}$$

Both $N$ and $Y$ are affine in $(c_1, s_1)$. Substitution of (18) into $s_3^2 + c_3^2 = 1$ gives

$$4a_2^2(Y - d_2)^2 + (N + K - 2d_2 Y)^2 = 4a_2^2 d_4^2. \tag{20}$$

Introduce the half-angle variable $t = \tan(q_1/2)$, so that

$$D = 1 + t^2, \qquad c_1 = \frac{1 - t^2}{D}, \qquad s_1 = \frac{2t}{D}. \tag{21}$$

Writing $Y = \widetilde{Y}/D$ and $N = \widetilde{N}/D$, their quadratic numerators are

$$\begin{aligned} \widetilde{Y}(t) &= c_\gamma\Big[y_w(1 - t^2) - 2x_w t\Big] + \tfrac{1}{2}hD, \\ \widetilde{N}(t) &= N_0 D - 2a_1\Big[x_w(1 - t^2) + 2y_w t\Big]. \end{aligned} \tag{22}$$

Multiplying (20) by $D^2$ gives

$$\boxed{P(t) = 4a_2^2(\widetilde{Y} - d_2 D)^2 + (\widetilde{N} + KD - 2d_2\widetilde{Y})^2 - 4a_2^2 d_4^2 D^2 = 0.} \tag{23}$$

Each quantity being squared is quadratic in $t$, so $P$ has degree at most four. Every real root supplies a candidate $q_1 = 2\arctan t$. The substitution $t = \tan(q_1/2)$ is undefined at $q_1 = \pi$, so this angle is not represented by any finite root of $P(t)$. To avoid missing a solution, we test $q_1 = \pi$ separately in (20) and retain it only if that equation is satisfied.

**4. Recover $q_2$ and $q_3$ for each candidate $q_1$.** For each candidate $q_1$, evaluate (15) and (18), then define $A = a_2 + d_4 c_3$. The remaining components of (16) are

$$X = c_2 A + s_2 d_3, \qquad Z = -s_2 A + c_2 d_3. \tag{24}$$

Thus

$$\begin{aligned} C_2 &= AX + d_3 Z = (A^2 + d_3^2)c_2, \\ S_2 &= d_3 X - AZ = (A^2 + d_3^2)s_2, \end{aligned} \qquad q_2 = \operatorname{atan2}(S_2, C_2). \tag{25}$$

The common factor is positive, so no division is needed before applying atan2. The elbow angle is recovered as $q_3 = \operatorname{atan2}(s_3, c_3)$. Since $A$ uses $c_3$ directly, $q_2$ and $q_3$ can be computed in either order.

If leading polynomial coefficients vanish, the degree reduces. For this robot, $P \equiv 0$ occurs when the target wrist centre is a reachable point on the base rotation axis. At such a position, $q_1$ is arbitrary, yielding a continuous family of arm configurations.

### Practical evaluation

Given $(R_d, p_d)$, compute $p_w$ from (14), followed by $h$, $K$ and $N_0$, and form $P$ using (22)–(23). For $P \not\equiv 0$, collect $q_1 = 2\arctan t$ from every real root and test $q_1 = \pi$ separately

in (20); for $P \equiv 0$, choose any $q_1$, for example 0, to represent the continuous family. For each candidate, evaluate $(X, Y, Z)$ from (15), set $N = X^2 + Y^2 + Z^2$, obtain $(s_3, c_3)$ from (18), and set $A = a_2 + d_4 c_3$. Then compute

$$q_2 = \operatorname{atan2}(d_3 X - AZ,\ AX + d_3 Z), \qquad q_3 = \operatorname{atan2}(s_3, c_3).$$

Check each triple against the target wrist position using forward kinematics, represent angles in $[-\pi, \pi]$ and remove duplicates modulo $2\pi$. Pass the valid arm configurations to the wrist-orientation problem.

## 3.2 Endpoint orientation

For each arm configuration $(q_1, q_2, q_3)$ found in the positioning step, we now determine the wrist angles $(q_4, q_5, q_6)$ that give the desired orientation $R_d$. Different arm configurations generally require different wrist angles, so this calculation is repeated for each arm solution.

### Derivation

**1. Express the desired orientation relative to the arm.** Using the same URDF link frames as in the positioning derivation, the orientation of `link_3_d3` relative to `base_link` is

$$R_{\text{arm}} = R_z(q_1) R_x(\gamma) R_y(q_2) R_z(q_3), \qquad \gamma = \pi/6. \tag{26}$$

The wrist joints rotate about their local $x$, $z$ and $x$ axes, respectively. There are no fixed rotations between these URDF link frames. Thus

$$R_d = R_{\text{arm}} R_{\text{wrist}}, \qquad R_{\text{wrist}} := R_{\text{arm}}^T R_d = R_x(q_4) R_z(q_5) R_x(q_6). \tag{27}$$

Here the transpose undoes the known arm rotation. Let $r_{ij}$ denote the entry in row $i$ and column $j$ of $R_{\text{wrist}}$.

**2. Use the first column to recover $q_5$ and $q_4$.** Since $R_x(q_6) e_x = e_x$, the first column is

$$R_{\text{wrist}} e_x = R_x(q_4) R_z(q_5) e_x = \begin{pmatrix} c_5 & c_4 s_5 & s_4 s_5 \end{pmatrix}^T = \begin{pmatrix} r_{11} & r_{21} & r_{31} \end{pmatrix}^T. \tag{28}$$

Thus $c_5 = r_{11}$. Using $c_4^2 + s_4^2 = 1$ also gives $\rho := \sqrt{r_{21}^2 + r_{31}^2} = |s_5|$. For $\rho > 0$, take each sign $s_5 = \pm\rho$. Dividing the last two entries by $s_5$ gives $c_4$ and $s_4$, so the two angle pairs are

$$q_5 = \operatorname{atan2}(s_5, r_{11}), \qquad q_4 = \operatorname{atan2}\left(\frac{r_{31}}{s_5}, \frac{r_{21}}{s_5}\right). \tag{29}$$

Keep the chosen sign of $s_5$ in both fractions. The case $\rho = 0$ is treated in step 4.

**3. Undo $q_4$ to recover $q_6$.** Once $q_4$ is known, multiply on the left by $R_x(-q_4)$ to undo its rotation:

$$B := R_x(-q_4) R_{\text{wrist}} = R_z(q_5) R_x(q_6). \tag{30}$$

The third row of $R_z(q_5)$ is $(0, 0, 1)$, so the third row of $B$ is $(0, s_6, c_6)$. We can therefore recover $q_6$ without dividing by $s_5$:

$$q_6 = \operatorname{atan2}(B_{32}, B_{33}). \tag{31}$$

When $\rho > 0$, the two wrist solutions are related by $(q_4, q_5, q_6) \mapsto (q_4 + \pi, -q_5, q_6 + \pi)$, with all angles taken modulo $2\pi$.

**4. Handle the cases where $s_5 = 0$.** When $\rho = 0$, we have $q_5 = 0$ or $q_5 = \pi$ modulo $2\pi$. The formula for $q_4$ divides by $s_5$ and no longer applies. The axes of joints 4 and 6 lie on the same line, so their angles cannot be determined separately. The wrist rotation reduces to $R_x(q_4 + q_6)$ for $q_5 = 0$ and $R_x(q_4 - q_6)R_z(\pi)$ for $q_5 = \pi$. Thus only a sum or a difference is fixed:

$$\begin{aligned} q_5 = 0 \ (r_{11} = 1): &\quad q_4 + q_6 = \sigma := \operatorname{atan2}(r_{32}, r_{22}), \\ q_5 = \pi \ (r_{11} = -1): &\quad q_4 - q_6 = \delta := \operatorname{atan2}(-r_{32}, -r_{22}). \end{aligned} \tag{32}$$

These relations hold modulo $2\pi$. We may choose any $q_4$ and set $q_6 = \sigma - q_4$ in the first case or $q_6 = q_4 - \delta$ in the second. Choosing $q_4 = 0$ gives the representatives $(0, 0, \sigma)$ and $(0, \pi, -\delta)$ for $(q_4, q_5, q_6)$.

### Practical evaluation

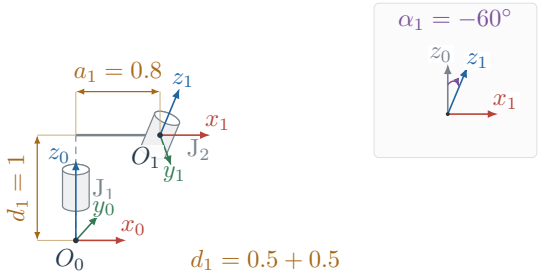
For each arm solution, use both wrist branches from steps 2–3 when $\rho$ exceeds a small numerical tolerance. Otherwise, take $q_4 = 0$ and use the sign of $r_{11}$ to select the case in (32). Combine each wrist solution with its arm solution, represent all angles in $[-\pi, \pi]$, and remove duplicates modulo $2\pi$. Verify the position and orientation against $(p_d, R_d)$ by forward kinematics, using separate tolerances.

# A DH frame constructions

Figure 2 collects all six frame-to-frame constructions at the home configuration ($q = 0$), using one fixed viewing direction and length scale. Lengths are in metres; angles are shown in degrees for readability.

**1 Base to shoulder:** $\{0\} \to \{1\}$

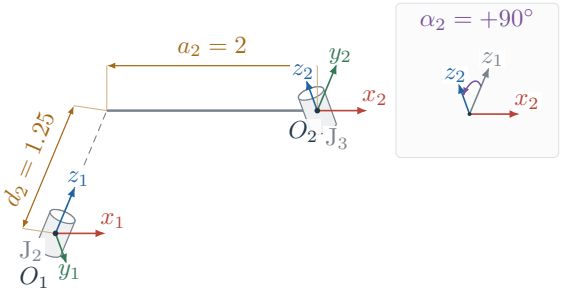
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$




The full base height is 1 m. The 30° URDF roll of joint 2 gives $\alpha_1 = 30° - 90° = -60°$.

**2 Shoulder to elbow:** $\{1\} \to \{2\}$

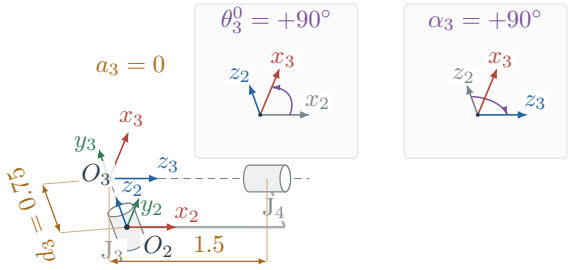
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$




The $x$ axes remain parallel. The offset along $z_1$ and the common normal along $x_2$ locate $O_2$ on joint 3.

**3 Axis intersection:** $\{2\} \to \{3\}$

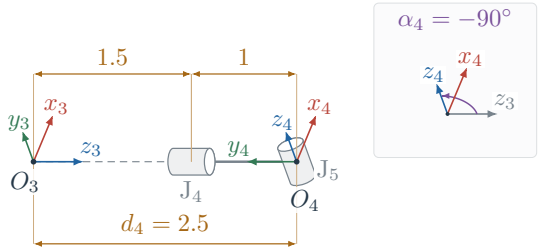
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$




$O_3$ is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed +90° offset in $\theta_3$.

**4 Reach the wrist:** $\{3\} \to \{4\}$

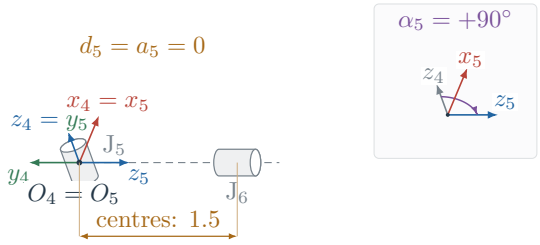
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$




The distance starts at $O_3$, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

**5 Coincident origins:** $\{4\} \to \{5\}$

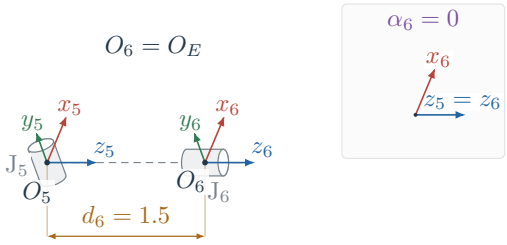
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$




The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

**6 Wrist to endpoint:** $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$




At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

**Figure 2:** Standard DH frame assignment. Cylinders denote physical revolute joints $J_i$; dots mark DH origins $O_i$. $x$, $y$ and $z$ axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Boxed angle sketches show the rotation $x_2 \to x_3$ about $+z_2$, giving $\theta_3^0$, and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.